

Basic Diagnostics Specification

Common Electrical Specification

Basic Diagnostics Specification

Range Of Validity

Vinfast_VF8_9_Premium

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1 General Information

Req ID	277	VinFast ID		Version number	2	Requirement Type	Heading
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1.1 Language

Req ID	278	VinFast ID		Version number	2	Requirement Type	Heading
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This specification is written in english.

1.2 Change History

Req ID	279	VinFast ID		Version number	17	Requirement Type	Heading
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Version	Req.ID. (Req. Type)	Brief Description	Date	Responsible person
3.0.0	all	Final Versio number	08.05.2018	Lutz.Neubacher
0.1	all	Carry over the version 3.0 of VF35 project	15.03.2022	Amy.Zhang
1.4	Req ID 279	Delete some redundant description information	15.05.2022	Perry.peng
1.5	Req ID 290	Modified the conditions for diagnostic	10.07.2022	Perry.peng
	Req ID 291	Changed Diagnostic Trouble Codes to Fault Mananement	10.07.2022	Perry.peng
	Req ID 306	Redefined the dlc of the CANFD	22.07.2022	Perry.peng
	Req ID 336	Optimized the usage of dignostic service	22.07.2022	Perry.peng
	Req ID 27014	Optimized the dignostic session and addressing	22.07.2022	Perry.peng
	Req ID 26511	Added some contents of Dolp	22.07.2022	Perry.peng
	Req ID 335	Change ECU-programming security level O into M	18.08.2022	Perry.peng

Table 1: Change History

2 Introduction

Req ID	281	VinFast ID		Version number	2	Requirement Type	Heading
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2.1 Scope

Req ID	282	VinFast ID		Version number	2	Requirement Type	Heading
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This document is the basic document for diagnostic task in the current project and describes the main diagnostic requirements:

- Basic Diagnostic functionalities
- Basic ON-Board diagnostic functionality
- Basic OFF-Board diagnostic requirements
- Additional featuresLinked specifications and documentations

Also the following specification follow this requirement:

- CES_Basic DID/PID Definition
- CES_Coding_Specification
- CES_Flash Specification

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2.2 Availability

Req ID	283	VinFast ID		Version number	2	Requirement Type	Heading
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The focus of this document is on the last level of technical standards and the level of development.

2.3 Terms and Definitions

Req ID	284	VinFast ID		Version number	2	Requirement Type	Heading
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2.3.1 OFF-Board communication

Req ID	285	VinFast ID		Version number	1	Requirement Type	Heading
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OFF-Board communication includes all processes of data communication between a vehicle and the tester device. The OFF-Board diagnostic communication is subject to special regulations of OBDII ISO 15031 and, SAE J1979 / SAE J17979-DA. All processes of this communication must be conforming to the requirements of Unified Diagnostic Service (ISO 14229).

2.3.2 ON-Board communication

Req ID	286	VinFast ID		Version number	1	Requirement Type	Heading
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ON-Board communication describes the communication processes between the ECUs over the vehicle communication network.

This includes automatically the ON-Board diagnosis as an important point which allows self-diagnosis of electronic components and systems. That means that ECUs must be able to detect faults within itself. On-Board diagnosis make sure the information and functionality for the Off-Board diagnostic.

On-Board Diagnostics runs in three levels:

- Detection
- Conditioning
- Storing

Once a fault is detected, a DTC with the relevant information about the fault reason must be stored within the ECU as a result of the On-Board Diagnostic.

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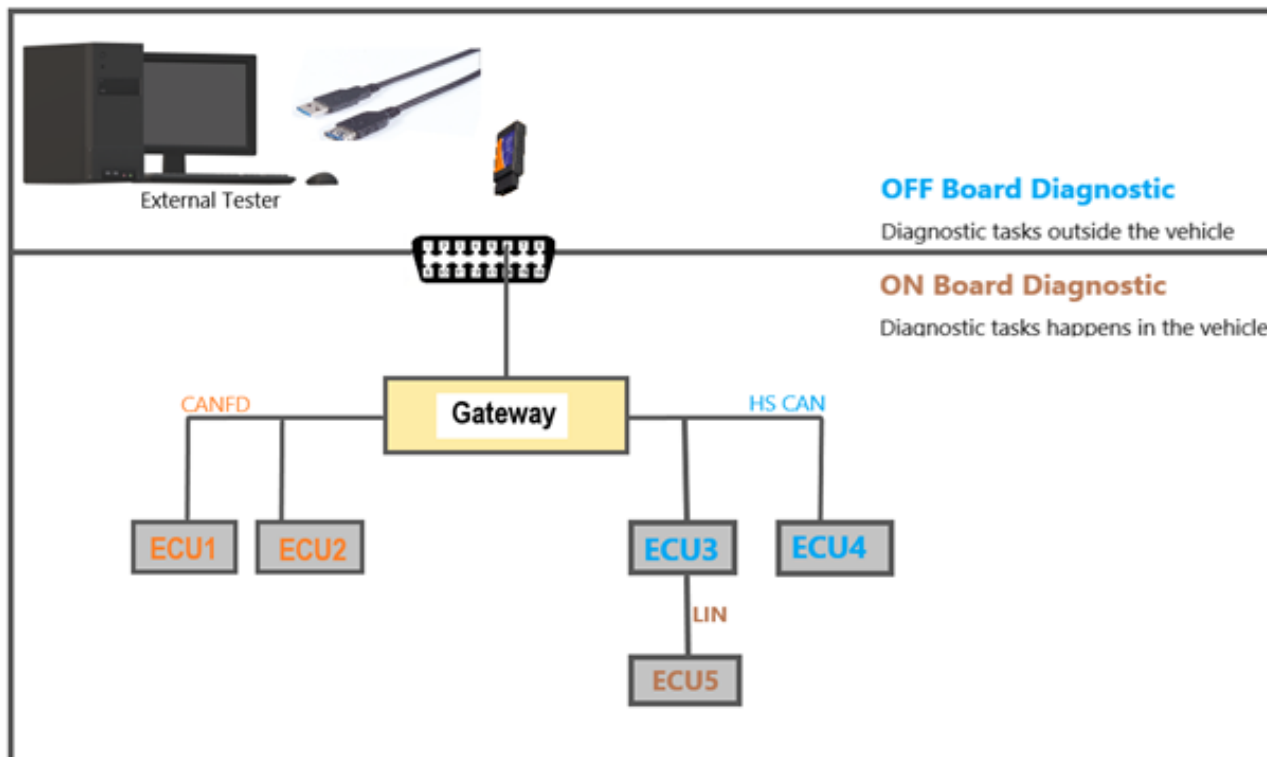


Figure 1: ON Board and OFF Board Diagnostic

3 Abbreviations

Req ID	287	VinFast ID		Version number	3	Requirement Type	Heading
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The following list of abbreviations are used throughout this document with the given meaning.

DTC	Diagnostics Trouble Code
DID	Data Identifier
PID	Parameter Identifier
CAN	Controller Area Network
LIN	Local Interconnect Network
CANFD	CAN with Flexible Data-Rate
Dolp	Diagnostic communication over Internet Protoco
ISO	International Organization for Standardization
OBDII	On Board Diagnostic
SAE	Society of Automotive Engineers
UDS	Unified Diagnostic Services
ODX	Open Diagnostic Data Exchange
SAE	Society of Automotive Engineers
FAC	Failure Attempt Counter
M	Mandatory
U	User conditional
C	Conditional
O	Optional

Table 2: Abbreviations

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4 Diagnostic Requirements Overview

Req ID	288	VinFast ID		Version number	2	Requirement Type	Heading
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4.1 Basic Requirements

Req ID	289	VinFast ID		Version number	5	Requirement Type	Heading
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The ECUs have to fulfil the requirements of the ISO 14229 UDS. A Gateway routes all necessary diagnostic commands and messages to the connected network components and has to provide the connection and communication to an OFF-Board Tester.

4.2 Conditions for Diagnosis

Req ID	290	VinFast ID		Version number	17	Requirement Type	Heading
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For Diagnostics task on the vehicle, basically requirements of vehicle safety and user safety must be complied.

Diagnostic requests from the Tester to the ECUs should be answered by the ECUs only in case of follow conditions:

- The voltage battery between 9V-16V
- The network communication is steady and reliable

If a deviation required (e.g. in Production), the necessary conditions must be agreed with the Diagnostic team.

For Flash/Programming tasks and Coding tasks additional conditions are necessary. Basically, a reprogramming or coding process should be allowed in follow situations:

- Electric motor speed is equal 0 rpm(tolerance ± 20 rpm), the signal please refer to DBC.
- Vehicle is standing (it means vehicle speed is less than 3km/h), the signal please refer to DBC.
- System voltage above 11.5V, the voltage can be the monitoring value of server or refer to DBC.

All other conditions, which can not have a negative impact on electronic components or generally functionality regarding vehicle safety.

4.3 Fault Mananement

Req ID	291	VinFast ID		Version number	5	Requirement Type	Heading
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The ECUs have to detect fault conditions internally and store Diagnostics Trouble Codes (DTC), if the fault condition is reached (Self diagnostic). Therefore internally the ECUs have to check e.g.the input and output for power supply, or the different signals and the CAN network.If required, also the ECU have to detect fault conditions on peripheral connected componentse.g. Sensors, Actuators, etc..

The DTCs have to be stored as 3 Byte Code for the UDS components. Therefore the Supplier of the ECUs have to follow the requirements of the ISO 15031 and actual SAE J2012.For detailed DTC range information please refer to document 'DTC_DID range for each component'.

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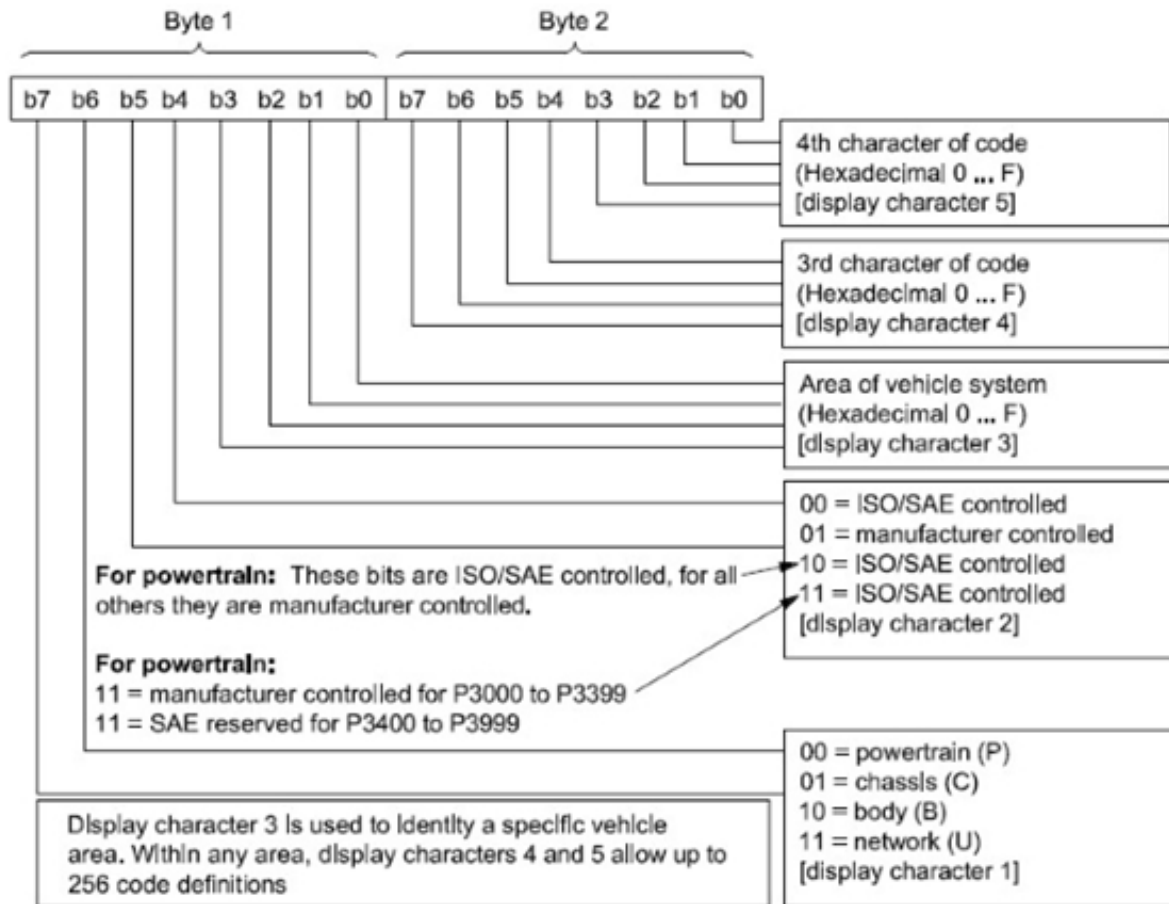


Figure 2: Structure of DTCs

Byte 3 Failure Type Byte definition please refer to 15031-6.

4.3.1 Snapshot data

Req ID	293	VinFast ID		Version number	5	Requirement Type	Distributed Function
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Every event memory entry comprises - in addition to the StatusOfDTC, the following information must be saved in non-volatile memory (NVM, e.g., EEPROM) when the fault occurs, and should be performed in the DTCSnapshotRecord:

- Mileage: the bit length and conversions, please refer to DBC.
- Vehicle speed: the bit length and conversions, please refer to DBC.
- Time stamp: the bit length and conversions, please refer to DBC.
- Battery voltage: the voltage can be the monitoring value of server or refer to DBC.
- Additional data to be defined by supplier on the system-specific basis.

All the relevant DIDs of snapshot data please refer to CES_Basic DID/PID Definition.

4.3.2 Extended data

Req ID	27218	VinFast ID		Version number	4	Requirement Type	Distributed Function
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Extended data information is a set of data which provides extended status information associated with a DTC; it contains fault occurrence counter, aged counter, aging counter and priority. The

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bytes required for extended data and the bytes sequence supported in "DTCExtendedDataRecord" of response message are defined in following table.

This extended record number is equal to 0x01.

Extended data	Fault occurrence counter	Aged counter	Aging counter	Priority
Cvt	M	M	M	M
Data length	1 Byte	1 Byte	1 Byte	4 bit
Byte position	Byte 0	Byte 1	Byte 2	Byte3(bit0~bit3)

Table 3: Global Extended Data Record

4.3.3 Aging counter

Req ID	295	VinFast ID		Version number	2	Requirement Type	Distributed Function
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Aging counter:

- The Aging counter should be 40dec and stored in the NVM when a fault first occurs.
- If an error-free detection cycle without a "test failed" in previous and current driving cycle, the Aging counter should be decremented by 1.
- The Aging counter must be reset to the initial value 40dec in the event of a "test failed" diagnostics result.
- If the Aging counter has reached the value 0, the event memory entry including the SnapShot should be erased.

4.3.4 Aged counter

Req ID	27219	VinFast ID		Version number	2	Requirement Type	Distributed Function
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This is a signal which indicates that a DTC has aged sufficiently to be removed from the ECU.If the DTC can't be deleted autonomously, ECU must save the aged counter status 255dec in the error memory.

4.3.5 Size of event memory

Req ID	297	VinFast ID		Version number	2	Requirement Type	Heading
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- The event memory with a minimum of 20 entries in a non-volatile memory is recommended.
- The event memory size must ensure enough for DTC and snapshot data storing.

4.4 Diagnostic Jobs

Req ID	298	VinFast ID		Version number	4	Requirement Type	Heading
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If required, the ECUs have to support services for read/write data and parameter, etc.. (e.g. HW number or SW number should be supported by all ECUs in the same data format. Production and Service have to define their special requirements together with logistic department and diagnostic department.)

4.5 Routines and Actuators

Req ID	299	VinFast ID		Version number	2	Requirement Type	Heading
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If it is required, a client can use a RoutineControl to trigger server's internal learning functions(e.g. Tire Pressure Learning, etc...), and use I/O control to substitute a value for an input signal, internal ECU function and/or control an output of an electronic system.(e.g.actuate relays or lamps, etc...)

4.6 Variant Coding

Req ID	300	VinFast ID		Version number	2	Requirement Type	Heading
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Servers that are codable should be coded by software and support different versions or functions of the ECU in this way.

4.7 Flash / Reprogramming

Req ID	301	VinFast ID		Version number	2	Requirement Type	Heading
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For faster and easier change of software during the development, production and after-sales, relevant ECUs should support Flash/Reprogramming functionality.

4.8 CAN/CANFD Diagnostic

Req ID	26512	VinFast ID		Version number	2	Requirement Type	Heading
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If CAN/CANFD Diagnostic function is supported, the communication regarding ISO 15765 、 ISO 11898、 SAE J1939 and ISO14229 must be implemented.

4.9 LIN Diagnostic

Req ID	302	VinFast ID		Version number	3	Requirement Type	Heading
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If LIN Diagnostic function is supported, the communication regarding ISO 17987 and ISO14229 must be implemented.

4.10 Dolp

Req ID	26511	VinFast ID		Version number	4	Requirement Type	Heading
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If Dolp function is supported, the communication regarding CES_Dolp Specification、 ISO 13400 and ISO14229 must be implemented.

4.11 OBDII /EOBD requirements

Req ID	29531	VinFast ID		Version number	2	Requirement Type	Heading
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In case of OBDII relevant tasks, the relevant ECUs has to support this regarding the actualspecification ISO 15031 and SAEJ1979 / SAE J1979-DA. Following services must implemented inthe OBD relevant ECUs regarding the requirements of the ISO.

OBD Service	Service Name	Usage
\$01	Request current powertrain diagnostic data	M

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\$02	Request powertrain freeze frame data	M
\$03	Request emission-related diagnostic trouble codes	M
\$04	Clear/reset emission-related diagnostic information	M
\$05	Request oxygen sensor monitoring test results	N/A
\$06	Request on-board monitoring test results for specific monitored systems	M
\$07	Request emission-related diagnostic trouble codes detected during current or last completed driving cycle	M
\$08	Request control of on-board system, test or component	M
\$09	Request vehicle information	M
\$0A	Request emission-related diagnostic trouble codes with permanent status	M

Table 4: OBDII service

Note: Electric vehicle not applicable.

4.12 Supplier specific adjustments

Req ID	303	VinFast ID		Version number	2	Requirement Type	Heading
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The ECUs support additional requirements, based on the specification from Production Team and Service Team.

Some jobs will happen in the plant (e.g. calibration, coding, learning process...), in order to support production, the supplier must provide the detail information for EOL.

The following requirement should be implemented:

1. methods to initialize the server.
2. adjustment or calibration like steering wheel calibration/ ESP calibration/ lifting window calibration, etc.
3. Variant coding.
4. Learning process like IMMO.
5. EOL sequences to ensure error-free operation.
6. Activation/Deactivation modes like "Transport Mode" (The definition of the transport mode refers to energy management specification).

The description can for example: consist of "reading measurement values", "reading general data", "writing data", "RoutineControl", "I/O Control" and so on.

If some function which is only designed for production purposes, effective locking mechanism must be have to ensure this function would not be implemented after production.

5 Diagnostic Communication

Req ID	304	VinFast ID		Version number	2	Requirement Type	Heading
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5.1 Communication Speed

Req ID	305	VinFast ID		Version number	5	Requirement Type	Heading
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The Diagnostic CAN Bus speed is 500kpbs, LIN Bus speed is 19.2kpbs, CANFD Bus speed is 2Mbps.

For ethernet bus speed refers to CES_Doip_Specification.

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5.2 CAN/CANFD Data link layer

Req ID	306	VinFast ID		Version number	6	Requirement Type	Distributed Function
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CAN Data link layer:

The data link layer shall meet the requirements of ISO11898-1.

The data length of all diagnostic request and response frames shall be 8 bytes, otherwise the ECU will ignore the diagnostic request frame.

When the diagnostic response length is less than 8 bytes, the empty bytes shall be padded with 0x00.

CANFD Data link layer:

All CANFD nodes must only support CANFD frame diagnostic requests and ignore CAN frame diagnostic requests.

The diagnostic request is CANFD frame, the response frame of CAN FD nodes shall be CANFD frame.

The CAN_DL of diagnostic request frame shall always set based on the TAB:Classical CAN/CAN FD Data Length value; When the data length of all diagnostic request and response is less than CAN_DL, the empty bytes shall be padded with 0x00, otherwise the frame shall be ignored.

Note:

Gateway should do the CAN to CANFD, CANFD to CAN transition between Client and Server.

Data Length Code (DLC)	Classical CAN Data Length (CAN_DL)	CAN FD Data Length (CAN_DL)
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	8	12
10	8	16
11	8	20
12	8	24
13	8	32
14	8	48
15	8	64

Table 5: Classical CAN/CAN FD Data Length value

5.3 Network Layer Parameters

Req ID	26515	VinFast ID		Version number	3	Requirement Type	Distributed Function
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Parameter	Timeout value	Performance Requirement
N_As/N_Ar	70ms	--
N_Bs	150	--
N_Br	--	(N_Br+ N_Ar)<70ms
N_Cs	--	(N_Cs+ N_As)<70ms
N_Cr	150ms	--
Parameter	Application mode	Bootloader mode
BS	8	0
STmin	20ms	0

Table 6: Requirement for Network Layer

5.4 Diagnostic Layer Parameters

Req ID	313	VinFast ID		Version number	5	Requirement Type	Distributed Function
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Parameter	Min	Max	Timeout	Unit
P2 _{server}	0	50	n/a	ms
P2 _{client}	n/a	n/a	150	ms
P2* _{server}	0	5000	n/a	ms
P2* _{client}	n/a	n/a	5000	ms
P3 _{client_phys}	P2 _{server}	n/a	n/a	ms
P3 _{client_func}	P2 _{server_max}	n/a	n/a	ms

Table 7: Requirement of Diagnostic Layer Parameter1

Parameter	Min	Nominal	Timeout	Unit
S3 _{server}	n/a	n/a	5000	ms
S3 _{client}	0	2000	4000	ms

Table 8: Requirement of Diagnostic Layer Parameter2

Note: for emission related system (ISO-15765-4) the according timings could to be applied.

5.5 Diagnostic Layer Error Handling

Req ID	315	VinFast ID		Version number	2	Requirement Type	Heading
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Error handling for the application and session layer to be fulfilled by the client and server regardless of the addressing method, shall be in accordance with the ISO 15765- 3. The client may repeat a UDS request a maximum of two times. Therefore, a UDS message is sent to one or more servers a total of three times.

5.6 Addressing method

Req ID	316	VinFast ID		Version number	2	Requirement Type	Heading
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Following addressing method is required:

- Normal Addressing

For this method the requirements of the ISO-15765 is binding. The using of the method is described in the ISO-14229 for UDS ECUs. Deviations must be coordinated with the Project Architecture and Diagnostic Team. For further information refer the document:

- CES_ECU Identifier Definition

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5.7 ECU Address

Req ID	317	VinFast ID		Version number	2	Requirement Type	Heading
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Each service can be transmitted to one specific ECU with physical addressing, to several ECUs or to all systems in a bus system with functional addressing.

A service to all systems is called a broadcast service.

For Services that support physical or functional addressing, refer to TAB:Supported diagnostic services of ECU.

For Diagnostic request and response IDs, refer the document:

- CES_ECU Identifier Definition.

Deviations shall agree by the Project department.

5.8 Diagnostic Message Format

Req ID	28129	VinFast ID		Version number	1	Requirement Type	Distributed Function
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Message Type	CAN ID	CAN Data Field							
		Byte 0		Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte n
		Bit 7-4	Bit 3-0						
Single Frame CAN_DL=8 ^[1]	CAN ID	0000	SF_DL	Data					
Single Frame CAN_DL>8 ^[2]	CAN ID	0000	0000	SF_DL	Data				
First Frame FF_DL≤4095 ^[3]	CAN ID	0001	FF_DL	FF_DL	Data				
First Frame FF_DL>4095 ^[4]	CAN ID	0001	0000	0000 0000	FF_DL	FF_DL	FF_DL	FF_DL	Data
Consecutive Frame ^[5]	CAN ID	0010	SN	Data					
Flow Control	CAN ID	0011	FS	BS	STmin	Padding			
1 In single frame transmission, CAN DL=8 traditional CAN frame adopts traditional PCI format. 2 In single frame transmission, CAN DL=64 CAN FD frame adopts new PCI format, that is, the original SF DL is set to 0 (invalid value), and the whole byte 1 is used to represent SF DL. 3.Only when FF_DL is more than 64, it can be used with multi-frame transmission. 4 In multi-frame transmission, the actual FF_DL ≤ 4095, the first frame is sent in the traditional PCI format. 5 In the multi-frame transmission process, the actual FF_DL > 4095, the new PCI format is used to send the first frame, i.e., the original FF_DL is set to 0 (invalid value), and the byte 2-5 is used to represent FF_DL.									

Table 9: Diagnostic Message Format

6 System Description

Req ID	318	VinFast ID		Version number	2	Requirement Type	Heading
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6.1 General System Description

Req ID	319	VinFast ID		Version number	2	Requirement Type	Heading
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For the specific BUS architecture, refer the document:

- System schematic

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6.2 CAN/CANFD Network

Req ID	320	VinFast ID		Version number	2	Requirement Type	Heading
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For the requirement of CAN/CANFD network, refer to the document:

- CES_CAN High Speed Specification
- CES_CAN FD Specification

6.3 LIN Network

Req ID	321	VinFast ID		Version number	2	Requirement Type	Heading
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For the requirement of LIN network, refer to the document:

- CES_LIN Specification

6.4 Specific Requirements

Req ID	322	VinFast ID		Version number	2	Requirement Type	Heading
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For Common DIDs, refer the document:

- CES_Basic DID/PID Definition

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Req ID	323	VinFast ID		Version number	2	Requirement Type	Heading
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7.1 Diagnostic Protocol

Req ID	324	VinFast ID		Version number	5	Requirement Type	Heading
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The communication between vehicle and tester must comply with the requirements of Unified Diagnostic Services(ISO 14229).

7.2 Diagnostic Data

Req ID	325	VinFast ID		Version number	4	Requirement Type	Heading
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The defined data exchange format for all ECUs is an ODX package. This data package has to include all necessary diagnostic information of the ECUs. The Supplier has to deliver a fully tested file regarding ODX authoring guideline and ODX standard ISO 22901 as Version 2.2.0 The Supplier makes sure, that the file is running and can be used on a valid MCD Server regarding ISO 22900.

All data mentioned in this requirement are hexadecimal unless otherwise specified.

7.3 Diagnostic Sessions

Req ID	326	VinFast ID		Version number	2	Requirement Type	Heading
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For diagnostic task, the ECU must support different sessions. This chapter introduced the different diagnostic sessions, which should be available for the relevant ECUs.

7.3.1 Default Session(0x01)

Req ID	327	VinFast ID		Version number	2	Requirement Type	Heading
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The Default Session shall be enabled, if no other session is requested after Ignition OFF-ON/ ECU Reset.

7.3.2 Programming Session(0x02)

Req ID	328	VinFast ID		Version number	2	Requirement Type	Heading
--------	-----	------------	--	----------------	---	------------------	---------

The Programming Session shall be enabled, if the Flash/Re-Programming sequence is requested by the tester.

7.3.3 Extended Session(0x03)

Req ID	329	VinFast ID		Version number	2	Requirement Type	Heading
--------	-----	------------	--	----------------	---	------------------	---------

The Extended Session shall be enabled, if specific diagnostic tasks requested by the tester.

7.3.4 Coding Session(0x41)

Req ID	330	VinFast ID		Version number	2	Requirement Type	Heading
--------	-----	------------	--	----------------	---	------------------	---------

The Coding Session shall be enabled, if a Coding process is requested by the tester.

7.3.5 Session Transition

Req ID	331	VinFast ID		Version number	3	Requirement Type	Heading
--------	-----	------------	--	----------------	---	------------------	---------

Basic Diagnostics Specification

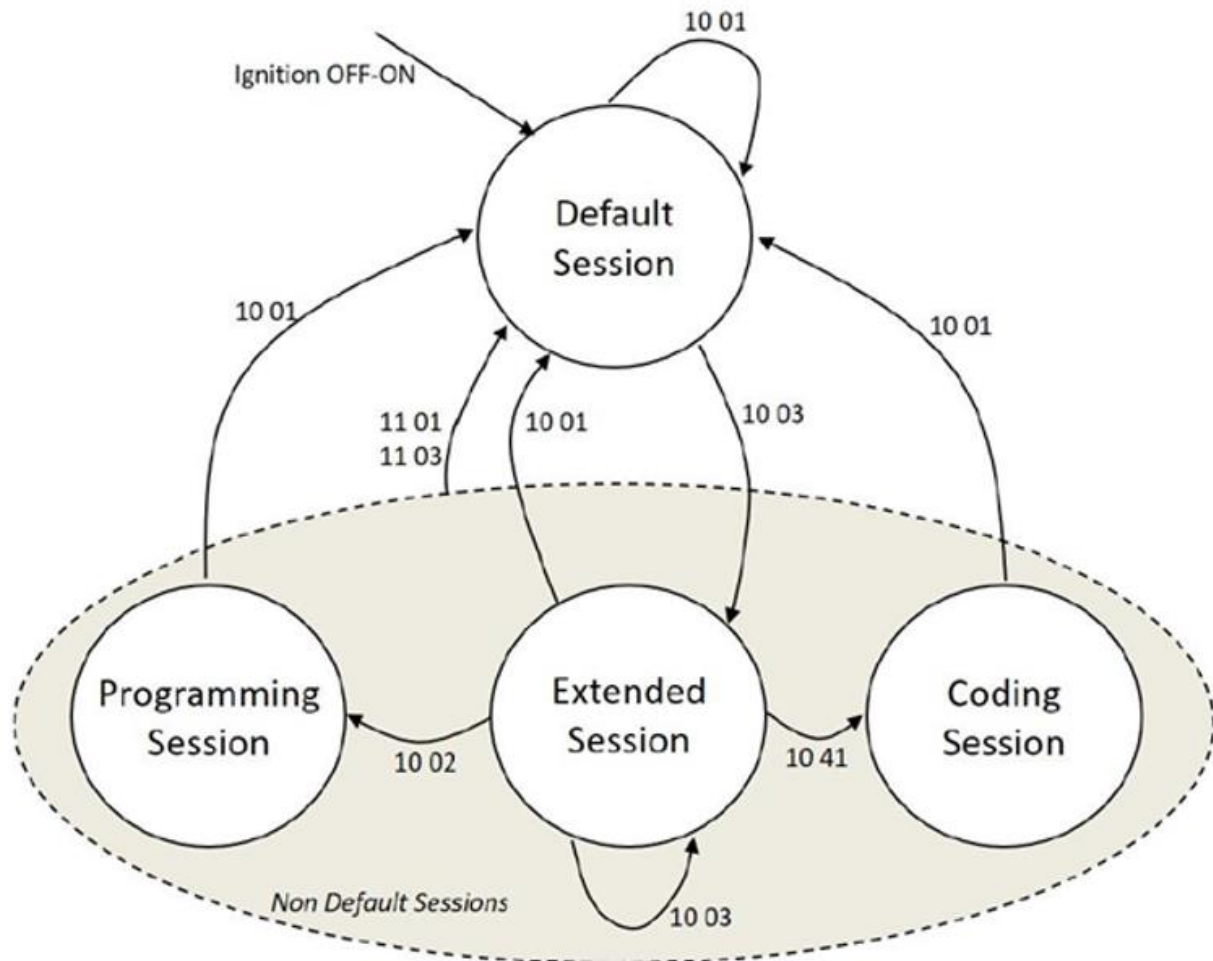


Figure 3: Session Transition Diagram

For space reasons, requests and responses are only briefly demonstrated, for actual DLC, please refer to chapter "CAN/CANFD Data link layer".

7.4 Security Access

Req ID	332	VinFast ID		Version number	2	Requirement Type	Distributed Function
--------	-----	------------	--	----------------	---	------------------	----------------------

ECU can support multiple security levels. diagnosis team defines four security levels(Level 0, Level 1, Level 2 and Level3). Each security level has an independent FAC(except Level 0) and a delay counter. Each security level has a Locked and Unlocked state. Only one security level can be Unlocked at a time.

The ECU shall record and save the FAC (Failure Attempt Counter). It is recommended to save the FAC to EEPROM. The saving time is determined by each ECU, but it must be ensured that the FAC will not be lost after the ECU is restarted (reset or powered on).

After power-on and reset, the ECU shall check the FAC of each security level first. If the FAC is less than 3, the ECU does not need to perform the security access delay. Otherwise, the ECU shall perform the security access delay (the delay time is 10s) and assign the FAC to 3.

The security access should follow the "Seed-Key" sequence, i.e., one "request Seed" should be sent by the tester, followed by one Key. If the correct key is sent by the tester, the corresponding security level enters the Unlocked state and the FAC is assigned 0. If the error key is sent by the tester, the FAC will increase by 1, and the ECU will respond with NRC 0x35. If the error key

Basic Diagnostics Specification

causes the FAC to be equal to 3, the ECU will repeat the 10s delay after replying to the negative response of the NRC 0x36.

If a security level is in a Locked state, the tester sends "request seed", the ECU should not send a seed with a value of 0. If the tester continuously sends "request seed ", the ECU shall return the same seed, but the FAC shall add 1. If the FAC is equal to 3 (i.e., the fourth continuous request seed) due to continuous transmission of " request seed ", the ECU shall return the negative response of NRC 0x37, and then perform 10s delay.

If the ECU receives the "request seed" request before the security access delay does not overflow, it shall reply to the negative response of NRC 0x37. After the security access is delayed overflow, the FAC is reduced by 1, and a security access is allowed.

If a certain security level is in the Unlocked state, if the ECU receives the "request seed" request, the ECU shall reply to the seed with a value of 0. At this time, if the tester continues to send the Key, the ECU shall reply to the negative response of NRC 0x24.

When the ECU is re-powered, reset, or a session jump is performed, the "Seed-Key" sequence will be reset.

The following figure shows ECU security access behavior:

Basic Diagnostics Specification

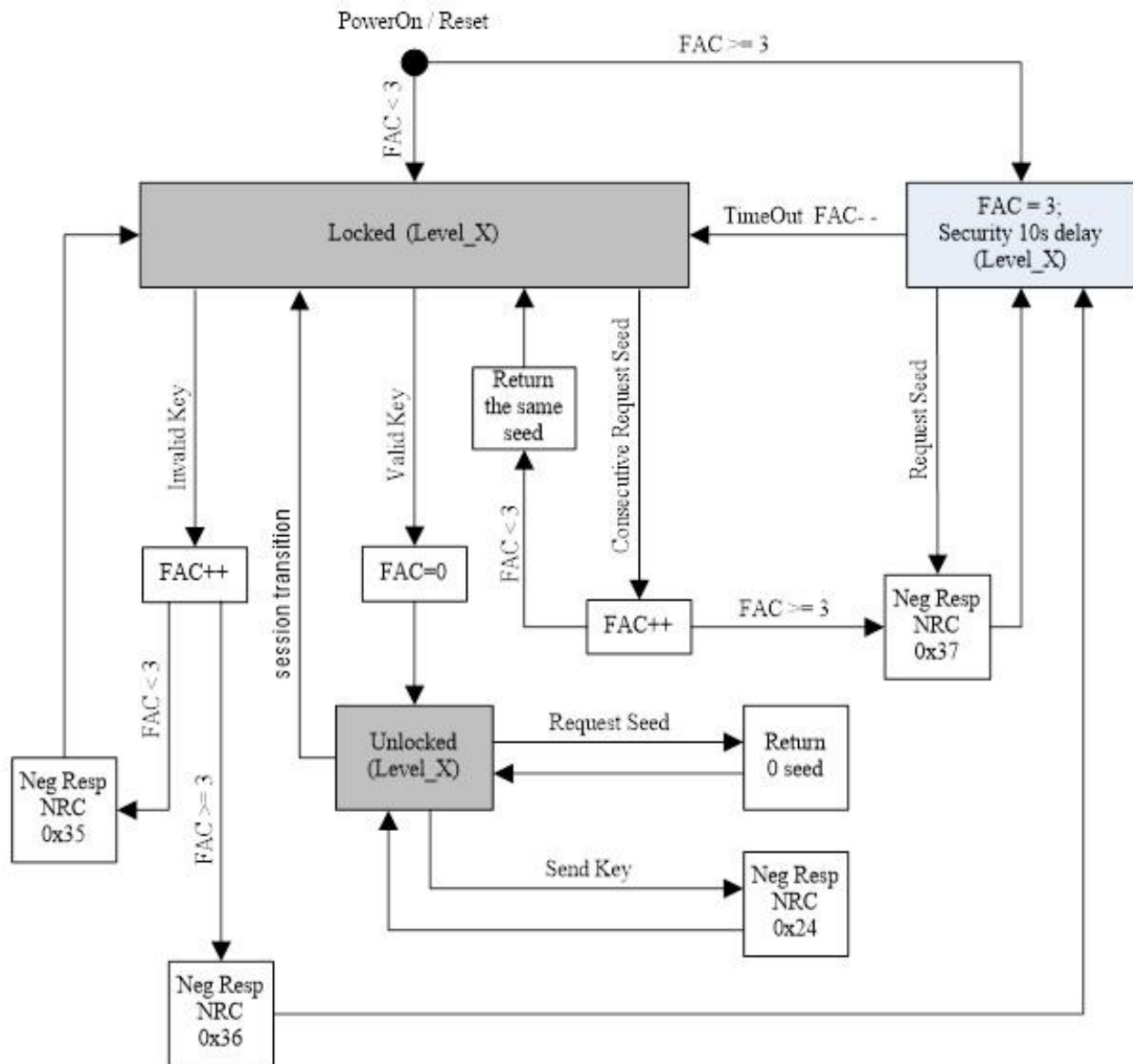


Figure 4: ECU security access behavior

7.4.1 Service sequences

Req ID	334	VinFast ID		Version number	3	Requirement Type	Heading
--------	-----	------------	--	----------------	---	------------------	---------

The following sequence is mandatory for unlocking locked services, parameters, and data:

1. The client requests a seed from the system.
2. The server sends the seed in the positive response.
3. The client sends the key to the server.
4. The server sends a confirmation:
 - Positive: The server confirms the correct key and unlocks all the services, parameters, or data protected with this SecurityLevel in the active NonDefaultSession.
 - Negative: The server rejects the key with a NegativeResponseCode.

Basic Diagnostics Specification

The Seed-Key algorithm for SecurityAccess refers to document: CES_Secure_Access Specification.

7.4.2 Security Levels

Req ID	335	VinFast ID		Version number	7	Requirement Type	Heading
--------	-----	------------	--	----------------	---	------------------	---------

Some data and services are protected against unauthorized execution using a SecurityLevel of Service 0x27.

Different security level will be used in different Scenarios.

The calculation result of the Key depends on the Random Number(RND) and the specific Key based on each level(OEM Key).

Security Level	Description	Key must be defined by	usage
0	Default Level	No Key needed	M
1	After Sales (Service)	RND and OEM Key	M
2	Manufacturing	RND and OEM Key	M
3	ECU-Coding	RND and OEM Key	O
4	ECU-Programming (Flash)	RND and OEM Key	M

Table 10: Security Level

Security Level	Description	RequestSeed	SendKey	Session	usage
0	Default Level	-	-	Default	M
1	After Sales (Service)	0x01	0x02	Extended	M
2	Manufacturing	0x03	0x04	Extended	M
3	ECU-Coding	0x05	0x06	Coding	O
4	ECU-Programming (Flash)	0x07	0x08	Programming	M

Table 11: Seed and Key Sub-Functions

7.5 Diagnostic Services

Req ID	336	VinFast ID		Version number	8	Requirement Type	Heading
--------	-----	------------	--	----------------	---	------------------	---------

This chapter describes the diagnostics services for the UDS protocol.

SID (Hex)	Service Name	Usage
10	DiagnosticSessionControl	M
11	EcuReset	M
27	SecurityAccess	M
28	Communication Control	M
3E	TesterPresent	M
85	ControlDTC Setting	M
22	ReadData ByIdentifier	M
2E	WriteData ByIdentifier	M
2F	InputOutputControlByIdentifier	U
14	ClearDiagnostic Information	M
19	ReadDTC Information	M
31	RoutineControl	M
34	Request DownLoad	M
36	TransferData	M
37	Request TransferExit	M
83	AccessTimingParameter	U

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84	SecuredDataTransmission	U
86	ResponseOnEvent	U
87	LinkControl	U
23	ReadMemoryByAddress	U
24	ReadScalingDataByIdentifier	U
2A	ReadDataByPeriodicIdentifier	U
2C	DynamicallyDefineDataIdentifier	U
3D	WriteMemoryByAddress	U

Table 12: Supported diagnostic services of ECU

- For other UDS services in ISO-14229 but not in the list are supplier optional, used as needed.

7.5.1 Diagnostic service details

Req ID	27014	VinFast ID		Version number	7	Requirement Type	Heading
--------	-------	------------	--	----------------	---	------------------	---------

Following table shows how all the services, which shall be supported by ECU supplier, are supported in different session and ways of addressing.

Diagnostic Services List		Session				Addressing	
SID (Hex)	Service Name	Default	Extended	Coding	Programming	Physical	Functional
10	DiagnosticSessionControl	x	x	x	x	x	x
11	EcuReset	x	x	x	x	x	x
27	SecurityAccess	-	x	x	x	x	-
28	CommunicationControl	-	x	-	-	x	x
3E	TesterPresent	x	x	x	x	x	x
85	ControlDTCSetting	-	x	-	-	x	x
22	ReadDataByIdentifier	x	x	x	x	x	-
2E	WriteDataByIdentifier	-	x	x	x	x	-
2F	InputOutputControlByIdentifier	-	x	-	-	x	-
14	ClearDiagnosticInformation	x	x	-	-	x	x
19	ReadDTCInformation	x	x	-	-	x	x
31	RoutineControl	-	x	x	x	x	-
34	RequestDownload	-	-	-	x	x	-
36	TransferData	-	-	-	x	x	-
37	RequestTransferExit	-	-	-	x	x	-

Table 13: Diagnostic services details

"x": Represent supported

"-": Represent not supported

7.5.2 Sub-function parameter definition

Req ID	27024	VinFast ID		Version number	4	Requirement Type	Heading
--------	-------	------------	--	----------------	---	------------------	---------

The sub-function parameter tells the diagnostic application which functionality is being requested. The sub-function parameter byte is divided into two parts (on bit level) as the following table:

Basic Diagnostics Specification

Bit	Parameter name
7	suppressPosRspMsgIndicationBit This 7th bit in the sub-function byte indicates whether a positive response message shall be suppressed by the server or not: '0' = FALSE, do not suppress a positive response message (a positive response message is required). '1' = TRUE, suppress response message (a positive response message shall not be sent; the server being addressed shall not send a positive response message).
6~0	Sub-function parameter value The bits 6~0 of the sub-function parameter contain the sub-function parameter value of the service(\$00~\$7F)

Table 14: sub-function parameter definition

Independent of the suppressPosRspMsgIndicationBit, negative response messages are sent by the server(s). What recommend is that when functional addressing used, the NRC such as 0x11, 0x12, 0x7E, 0x7F, 0x31 can also be suppressed, Other NRC should be sent to the client.

7.5.3 Services support SPR bit

Req ID	367	VinFast ID		Version number	4	Requirement Type	Heading
--------	-----	------------	--	----------------	---	------------------	---------

The services need to support suppressPositiveResponseBit (SPR) are showed in following table.

Diagnostic Services List		Support SPR bit
SID (Hex)	Service Name	
10	DiagnosticSessionControl	×
11	EcuReset	×
27	SecurityAccess	-
28	CommunicationControl	×
3E	TesterPresent	×
85	ControlDTCSetting	×
22	ReadDataByIdentifier	-
2E	WriteDataByIdentifier	-
14	ClearDiagnosticInformation	-
19	ReadDTCInformation	-
2F	InputOutputControlByIdentifier	-
31	RoutineControl	-
34	RequestDownload	-
36	TransferData	-
37	RequestTransferExit	-

Table 15: Services support SPR bit

"×": Represent supported

"-": Represent not supported

7.5.4 \$10 DiagnosticSessionControl

Req ID	337	VinFast ID		Version number	8	Requirement Type	Distributed Function
--------	-----	------------	--	----------------	---	------------------	----------------------

This service is used by the client to enable different diagnostic sessions in the server(s).

Diagnostic Session Control Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value

Basic Diagnostics Specification

1	Diagnostic Session Control Request Service Id	M	10
Sub Function = []			
Data Byte No	Parameter Name	Usage	Data Value
2	sub-function = [diagnosticSessionType]	M	00 - FF
	defaultSession	M	01 /81
	programmingSession	M	02 /82
	extendedDiagnosticSession	M	03 /83
	codingSession	O	41/C1

Table 16: DiagnosticSessionControl Service Request

Diagnostic Session Control Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	Diagnostic Session Control Response Service Id	M	50
Sub Function = []			
Data Byte No	Parameter Name	Usage	Data Value
2	sub-function = [diagnosticSessionType]	M	00 - FF
Session Parameter Record			
Data Byte No	Parameter Name	Usage	Data Value
	sessionParameterRecord[] #1 =	M	00 - FF
3	[P2Server_Max(high byte)	M	00 - FF
4	P2Server_Max(low byte)	M	00 - FF
5	P2*Server_Max(high byte)	M	00 - FF
6	P2*Server_Max(low byte)]	M	00 - FF

Table 17: DiagnosticSessionControl positive response

Parameters	Description	Bytes	Resolution
P2Server_Max	This value indicates the server's standard timing P2 in the active session.	2	1
P2*Server_Max	This value identifies the server's extended P2 timing which is started after an NRC 0x78 (RequestCorrectlyReceived-ResponsePending).	2	10

Table 18: SessionParameterRecord

The following example shows the change to the ExtendedSession. For the change to the ExtendedSession, the server returns the following parameters:

P2Server_Max = 0x0032 = 50 ms

P2*Server_Max = 0x00C8 = 2 000 ms

7.5.5 \$11 ECUReset

Req ID	339	VinFast ID		Version number	6	Requirement Type	Distributed Function
--------	-----	------------	--	----------------	---	------------------	----------------------

This service requests the server to effectively perform an ECU reset based on the content of the ResetType parameter value.

ECUReset Resquest Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	ECUReset Request Service Id	M	11
subFunction = []			
Data Byte No	Parameter Name	Usage	Data Value

Basic Diagnostics Specification

2	subFunction = [resetType]	M	00 - FF
	HardReset	M	01/81
	SoftReset	M	03/83

Table 19: ECUReset service request

ECUReset Response Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	ECUReset Response Service Id	M	51
ResetType			
Data Byte No	Parameter Name	Usage	Data Value
2	resetType	M	01/03

Table 20: ECUReset service positive response

7.5.6 \$27 SecurityAccess

Req ID	341	VinFast ID		Version number	8	Requirement Type	Distributed Function
--------	-----	------------	--	----------------	---	------------------	----------------------

This service enables to access data and services that are not normally allowed for security reasons.

ECUReset Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	SecurityAcces Request Service Id	M	27
subFunction = [securityAccessType = requestSeed]			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [securityAccessType = requestSeed]	M	00 – FF
	requestSeed (After Sale)	M	01
	requestSeed (Manufacturing)	M	03
	requestSeed (ECU-Coding)	U	05
	requestSeed (ECU-Programing)	M	07
subFunction = [securityAccessType = sendKey]			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [securityAccessType = sendKey]	M	00 – FF
	sendKey	M	02
	sendKey	M	04
	sendKey	M	06
	sendKey	M	08
3	securityKey[] = [key#1 (high byte)	M	00 - FF
:	:	:	:
#n	key#m (low byte)]	U	00 - FF

Table 21: SecurityAccess Service request

SecurityAccess Response Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	SecurityAccess Response Service Id	M	67
securityAccessType			
Data Byte No	Parameter Name	Usage	Data Value
2	securityAccessType	M	00 – 7F
securitySeed			
Data Byte No	Parameter Name	Usage	Data Value
3	securitySeed[] = [seed#1 (high byte)	M	00 – FF

Basic Diagnostics Specification

:	:	:	:
#n	seed#m (low byte)]	M	00 – FF

Table 22: SecurityAccess Positive response

7.5.7 \$28 CommunicationControl

Req ID	343	VinFast ID		Version number	5	Requirement Type	Distributed Function
--------	-----	------------	--	----------------	---	------------------	----------------------

This service is used to switch on and off the sending and/or receiving of certain messages of a system in order to increase the bandwidth of the respective bus system (via the bus application).

CommunicationControl Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	CommunicationControl Request Service Id	M	28
subFunction = [controlType]			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [controlType]	M	00 - FF
	enableRxAndTx	M	00 /80
	enableRxAndDisableTx	M	01 /81
	disableRxAndEnableTx	U	02 /82
	disableRxAndTx	U	03 /83
communicationType			
Data Byte No	Parameter Name	Usage	Data Value
3	communicationType	M	00 - FF
	normalCommunicationMessages	M	01
	networkManagementCommunicationMessages	M	02
	NormalandNetworkManagementCommunicationMessages	M	03
1, If an ECU implements network management function, 02 shall support this communication type for communicationControl service.			
2, For flashing function, 03 should be supported by all the ECUs.			

Table 23: CommunicationControl service request

CommunicationControl Response Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	CommunicationControl Response Service Id	M	68
controlType			
Data Byte No	Parameter Name	Usage	Data Value
2	controlType	M	00/01/02/03

Table 24: Positive response message

7.5.8 \$3E TesterPresent

Req ID	345	VinFast ID		Version number	5	Requirement Type	Distributed Function
--------	-----	------------	--	----------------	---	------------------	----------------------

This service shall be used to indicate to a server that the client is present and certain previously activated diagnostic functions must continue to be maintained.

The TesterPresent (3Ehex) must be used by the client to maintain a previously activated NonDefaultSession in one or more servers.

The valid TesterPresent (3Ehex) request with SuppressPositiveResponseBit == 1 and must be processed by the server in the bypass process.

Bypass process means that the request is not handled in the UDS service and does not generate a positive or negative response. The timer S3Server as per the "Diagnostic Layer Parameters" section is retrigged.

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TesterPresent Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	TesterPresent Request Service Id	M	3E
subFunction = [zeroSubFunction]			
Data Byte No	Parameter Name	Usage	Data Value
2	sub-function = [zeroSubFunction]	M	00 - 7F
	zeroSubFunction	M	00 /80

Table 25: TesterPresent service request

TesterPresent Response Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	TesterPresent Response Service Id	M	7E
zeroSubFunction			
Data Byte No	Parameter Name	Usage	Data Value
2	zeroSubFunction	M	00

Table 26: TesterPresent service positive response

7.5.9 \$85 ControlDTCSetting

Req ID	347	VinFast ID		Version number	5	Requirement Type	Distributed Function
--------	-----	------------	--	----------------	---	------------------	----------------------

This service is used to enable and disable the setting of Diagnostic Trouble Codes(DTCs) in the ECU(s).

ControlDTCSetting Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	ControlDTCSetting Request Service Id	M	85
subFunction = []			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [DTCSettingType]	M	00 - FF
	on	M	01 /81
	off	M	02 /82

Table 27: ControlDTCSetting service request message

ControlDTCSetting Response Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	ControlDTCSetting Response Service Id	M	C5
DTCSettingType			
Data Byte No	Parameter Name	Usage	Data Value
2	DTCSettingType	M	01 /02

Table 28: ControlDTCSetting service positive response

7.5.10 \$22 ReadDataByIdentifier

Req ID	349	VinFast ID		Version number	7	Requirement Type	Distributed Function
--------	-----	------------	--	----------------	---	------------------	----------------------

The request message requests data record values from the server by one or more dataIdentifier(s).

ReadDataByIdentifier Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	Read Data By Identifier Request Service Id	M	22
dataIdentifier[] #1 = []			
Data Byte No	Parameter Name	Usage	Data Value

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2	dataIdentifier[] #1 = [byte#1 (MSB)	M	00 - FF
3	byte#2]	M	00 - FF
:	:	:	:
n-1	dataIdentifier[] #m = [byte#1 (MSB)	U	00 - FF
n	byte#2]	U	00 - FF

Table 29: ReadDataByIdentifier service request message

ReadDataByIdentifier Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	Read Data By Identifier Response Service Id	M	62
dataIdentifier[] #1 = []			
Data Byte No	Parameter Name	Usage	Data Value
2	dataIdentifier[] #1 = [byte#1 (MSB)	M	00 - FF
3	byte#2]	M	00 - FF
dataRecord[] #1 = []			
Data Byte No	Parameter Name	Usage	Data Value
4	Data Record [] = [data#1 (MSB)	M	00 - FF
:	:	:	:
(k-1)+4	data#k]	U	00 - FF
:	:	:	:
dataIdentifier[] #m = []			
Data Byte No	Parameter Name	Usage	Data Value
n-(o-1)-2	dataIdentifier[] #m = [byte#1 (MSB)	U	00 - FF
:	:	:	:
n-(o-1)-1	byte#2]	U	00 - FF

Table 30: ReadDataByIdentifier service positive response

7.5.11 \$2E WriteDataByIdentifier

Req ID	351	VinFast ID		Version number	4	Requirement Type	Distributed Function
--------	-----	------------	--	----------------	---	------------------	----------------------

The client requests to write a record specified by a provided dataIdentifier.

WriteDataByIdentifier Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	WriteDataByIdentifier Request Service Id	M	2E
dataIdentifier[]			
Data Byte No	Parameter Name	Usage	Data Value
2	dataIdentifier[] = [byte#1 (MSB)	M	00 - FF
3	byte#2]	M	00 - FF
dataRecord[]			
Data Byte No	Parameter Name	Usage	Data Value
4	dataRecord[] = [data#1	M	00 - FF
:	:	:	:
m+3	data#m]	U	00 - FF

Table 31: WriteDataByIdentifier Service Request

WriteDataByIdentifier Response Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	WriteDataByIdentifier Response Service Id	M	6E
dataIdentifier[]			
Data Byte No	Parameter Name	Usage	Data Value

Basic Diagnostics Specification

2	dataIdentifier[] = [byte#1 (MSB)	M	00 - FF
3	byte#2]	M	00 - FF

Table 32: WriteDataByIdentifier Positive Response

7.5.12 \$14 ClearDiagnosticInformation

Req ID	353	VinFast ID		Version number	4	Requirement Type	Distributed Function
--------	-----	------------	--	----------------	---	------------------	----------------------

Client to erase the entire contents of the event memory or particular stored events. Additional information such as ambient data is also erased using this service.

The DTCs stored in event memory shall be cleared before the positive response is sent.

A subsequent call to read out the fault memory shall not retrieve any DTC which has been set before the last execution of ClearDiagnosticInformation.

Even if no DTC was stored, the ECU shall return with a positive response.

There shall be no sequence dependency to any other service. Even if the fault memory was not read, it may be cleared.

Clear Diagnostic Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	ClearDiagnosticInformation Request Service Id	M	14
groupOfDTC[]			
Data Byte No	Parameter Name	Usage	Data Value
2	groupOfDTC[] = [groupOfDTCHighByte	M	00 - FF
3	groupOfDTCMiddleByte	M	00 - FF
4	groupOfDTCLowByte]	M	00 - FF

Table 33: ClearDiagnosticInformation Service Request

Clear Diagnostic Information Positive Response Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	ClearDiagnosticInformation Positive Response Service Id	M	54

Table 34: ClearDiagnosticInformation Positive Response

7.5.13 \$19 ReadDTCInformation

Req ID	355	VinFast ID		Version number	9	Requirement Type	Distributed Function
--------	-----	------------	--	----------------	---	------------------	----------------------

Allows the client to request diagnostic information from the server (including DTCs, captured data, etc.)

Sub-function	Name	Usage
\$01	reportNumberOfDTCByStatusMask	M
\$02	reportDTCByStatusMask	M
\$03	reportDTCSnapshotIdentification	U
\$04	reportDTCSnapshotRecordByDTCNumber	M
\$05	reportDTCSnapshotRecordByRecordNumber	U
\$06	reportDTCExtendedDataRecordByDTCNumber	M
\$07	reportNumberOfDTCBySeverityMask	U
\$08	reportDTCBySeverityMaskRecord	U
\$09	reportSeverityInformationOfDTC	U
\$0A	reportSupportedDTC	M
\$0B	reportFirstTestFailedDTC	U
\$0C	reportFirstConfirmedDTC	U
\$0D	reportMostRecentTestFailedDTC	U

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\$0E	reportMostRecentConfirmedDTC	U
\$0F	reportMirrorMemoryDTCByStatusMask	U
\$10	reportMirrorMemoryDTCExtendedDataRecordByDTCNumber	U
\$11	reportNumberOfMirrorMemoryDTCByStatusMask	U
\$12	reportNumberOfEmissionsRelatedOBDDTCByStatusMask	U
\$13	reportEmissionsRelatedOBDDTCByStatusMask	U
\$14	reportDTCFaultDetectionCounter	U
\$15	reportDTCWithPermanentStatus	U

Table 35: Sub-functions of Service \$19

DTC status byte provides information on the status of the failure. The information is bit encoded with the full definitions for the individual bits provided in ISO 14229. The following table gives a summary of the meanings of the bits.

Bit	Bit field name	implement
7	warningIndicatorRequested	U
6	testNotCompletedThisMonitoringCycle	U
5	testFailedSinceLastClear	C2
4	testNotCompletedSinceLastClear	C2
3	confirmedDTC	M
2	pendingDTC	U
1	testFailedThisMonitoringCycle	C1
0	testFailed	M

Table 36: DTC Status byte bit definition

Note:

C1 bit1 is mandatory if bit2 is supported.bit1 is user optional if bit2 is not support.

C2 bit 4 and bit5 shall always be supported together.

7.5.14 \$2F InputOutputControlByIdentifier

Req ID	357	VinFast ID		Version number	5	Requirement Type	Distributed Function
--------	-----	------------	--	----------------	---	------------------	----------------------

This service is used by the tester to substitute a value for an input signal, internal ECU function and/or control an output of an electronic system referenced by an InputOutputDataIdentifier.

InputOutputControlByIdentifier Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	InputOutputControlByIdentifier Request Service Id	M	2F
dataIdentifier []			
Data Byte No	Parameter Name	Usage	Data Value
2	dataIdentifier#1[] = [byte#1 (MSB)	M	00 - FF
3	byte#2 (LSB)]	M	00 - FF
controlOptionRecord[]			
Data Byte No	Parameter Name	Usage	Data Value
4	controlOptionRecord#1[]	M	00 - FF
:	:	:	:
4+(m-1)	controlState#m]	C	00 - FF
controlEnableMaskRecord[]			
Data Byte No	Parameter Name	Usage	Data Value
4+m	controlEnableMaskRecord#1[] = [controlMask#1	C	00 - FF

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:	:	:	:
4+m+(r-1)	controlMask#r]	C	00 - FF

Table 37: inputOutputControlByIdentifier service request

InputOutputControlByIdentifier Response Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	InputOutputControlByIdentifier Response Service Id	M	6F
dataIdentifier []			
Data Byte No	Parameter Name	Usage	Data Value
2	dataIdentifier#1[] = [= [byte#1 (MSB)	M	00 - FF
3	byte#2 (LSB)]	M	00 - FF
controlStatusRecord[]			
Data Byte No	Parameter Name	Usage	Data Value
4	controlStatusRecord#1[] = [controlState#1/inputOutputControlParameter	C	00 - FF
:	:	:	:
4+(m-1)	controlState#m]	C	00 - FF

Table 38: inputOutputControlByIdentifier service positive response

For the inputOutputControlParameter, following parameters defined:

Hex	Description	Cvt
0x00	returnControlToECU	M
0x01	resetToDefault	U
0x02	freezeCurrentState	U
0x03	shortTermAdjustment	M

Table 39: inputOutputControl Parameter

7.5.15 \$31 RoutineControl

Req ID	359	VinFast ID		Version number	4	Requirement Type	Distributed Function
--------	-----	------------	--	----------------	---	------------------	----------------------

The client requests to start, stop a routine in the server(s) or requests the routine results.

RoutineControl Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	RoutineControl Request Service Id	M	31
sub-function = [routineControlType]			
Data Byte No	Parameter Name	Usage	Data Value
2	startRoutine	M	01
2	stopRoutine	M	02
2	requestRoutineResults	U	03
routineIdentifier []			
Data Byte No	Parameter Name	Usage	Data Value
3	routineIdentifier [] = [byte#1 (MSB)	M	00 - FF
4	byte#2]	M	00 - FF
routineControlOptionRecord[]			
Data Byte No	Parameter Name	Usage	Data Value
5	routineControlOptionRecord[] = [routineControlOption#1	C/U	00 - FF
:	:	:	:
n	routineControlOption#m]	C/U	00 - FF

Table 40: routineControl service request

Basic Diagnostics Specification

RoutineControl Response Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	RoutineControl Response Service Id	M	71
routineControlType			
Data Byte No	Parameter Name	Usage	Data Value
2	routineControlType	M	00 - FF
routineIdentifier []			
Data Byte No	Parameter Name	Usage	Data Value
3	routineIdentifier [] = [byte#1 (MSB)	M	00 - FF
4	byte#2]	M	00 - FF
routineStatusRecord[] = [routineStatus#1....routineStatus#m]			
Data Byte No	Parameter Name	Usage	Data Value
5	routineStatusRecord[] = [routineStatus#1	U	00 - FF
:	:	:	:
n	routineStatus#m	U	00 - FF

Table 41: routineControl service positive response

7.5.16 \$34 RequestDownload

Req ID	361	VinFast ID		Version number	3	Requirement Type	Distributed Function
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The client initialize the transfer of EEPROM or flash EEPROM data from the client to a server. Detailed information, please refer to the CES_Flash Specification.

7.5.17 \$36 TransferData

Req ID	363	VinFast ID		Version number	3	Requirement Type	Distributed Function
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This service is used to execute the previously initialized data transfer to the server. Detailed information, please refer to the CES_Flash Specification.

7.5.18 \$37 RequestTransferExit

Req ID	365	VinFast ID		Version number	3	Requirement Type	Distributed Function
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This service is used by the client to stop a data download. Detailed information, please refer to the CES_Flash Specification.

7.5.19 Other Services

Req ID	44145	VinFast ID		Version number	2	Requirement Type	Heading
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For other services, like \$83 AccessTimingParameter、\$84 SecuredDataTransmission、\$86 ResponseOnEvent、\$87 LinkControl、\$23 ReadMemoryByAddress、\$24 ReadScalingDataByIdentifier、\$2A ReadDataByPeriodicIdentifier、\$2C DynamicallyDefineDataIdentifier、\$3D WriteMemoryByAddress, the detailed information please refer to ISO 14229.

7.5.20 \$7F NegativeResponse

Req ID	27018	VinFast ID		Version number	6	Requirement Type	Heading
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Basic Diagnostics Specification

The negative response service shall be used by an ECU to indicate that a diagnostic service request message is either invalid, being terminated, or currently cannot be acted upon at the time of the request. The negative response code (NRC) list is as follows:

Hex Value	Response code	Mnemonic
10	generalReject	GR
11	serviceNotSupported	SNS
12	subFunctionNotSupported	SFNS
13	incorrectMessageLengthOrInvalidFormat	IMLOIF
14	responseTooLong	RTL
21	busyRepeatRequest	BRR
22	conditionsNotCorrect	CNC
24	requestSequenceError	RSE
31	requestOutOfRange	ROOR
33	securityAccessDenied	SAD
35	invalidKey	IK
36	exceedNumberOfAttempts	ENOA
37	requiredTimeDelayNotExpired	RTDNE
38 – 4F	ReservedByExtendedDataLinkSecurityDocument	RBEDLSD
72	generalProgrammingFailure	GPF
73	wrongBlockSequenceCounter	WBSC
78	requestCorrectlyReceived-ResponsePending	RCRRP
7E	subFunctionNotSupportedInActiveSession	SFNSIAS
7F	serviceNotSupportedInActiveSession	SNSIAS
92	VoltageTooHigh	VTH
93	voltageTooLow	VTL

Table 42: NRC List

8 Flash / Reprogramming

Req ID	369	VinFast ID		Version number	2	Requirement Type	Heading
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For details, refer to the document:

- CES_Flash Specification

9 Variant Coding

Req ID	370	VinFast ID		Version number	2	Requirement Type	Heading
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For details, refer to the document:

- CES_Coding_Specification

10 Appendix

Req ID	371	VinFast ID		Version number	2	Requirement Type	Heading
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10.1 DIDs / PIDs

Req ID	372	VinFast ID		Version number	2	Requirement Type	Heading
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10.1.1 Common Basic DIDs

Req ID	373	VinFast ID		Version number	2	Requirement Type	Heading
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For processes in Production and/or in Service it is necessary to use a number of common basic DIDs, which will be used for identification of components, software or the complete vehicle. The number and content of this common DIDs must be clarified together with the Logistic Team, Production and After-sales and should have the same structure for all used ECUs. Deviations have to be agreed with Diagnostic team.

A minimum of common Basic DIDs/PIDs are defined in the document:

- CES_Basic DID/PID Definition

10.1.2 ECU Specific DIDs/PIDs

Req ID	374	VinFast ID		Version number	4	Requirement Type	Heading
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All ECU specific Data/Parameter Identifier must be described in the respective component specification. And the range of specific DIDs/PIDs must follow the reference of DTC_DID_PID range for each component document.

10.2 Quality Ensure

Req ID	375	VinFast ID		Version number	3	Requirement Type	Heading
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The Supplier of the ECU has to deliver an Diagnostic Test Report.

The Supplier update the Report for any Function Release Step and send the full filled report to the Integration team during the relevant Function Delivery step.

The content of the Diagnostic Test Report has to filled completely and accurately by the Supplier.

11 Standards

Req ID	376	VinFast ID		Version number	2	Requirement Type	Heading
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11.1 Norms and Standards

Req ID	377	VinFast ID		Version number	3	Requirement Type	Heading
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Ref. No	Document Name
ISO14229	Road vehicles — Unified diagnostic services (UDS)
ISO15031	Road vehicles — Communication between vehicle and external equipment for emissions-related diagnostics
ISO15765	Road vehicles — Diagnostics on Controller Area Networks (CAN)
ISO22900	Road vehicles — Modular vehicle communication interface
ISO22901	Road vehicles — Open diagnostic data exchange (ODX)
SAE J2012	Diagnostic Trouble Code Definitions

Table 43: Normative references

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This Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

11.2 Additional Documents

Req ID	378	VinFast ID		Version number	4	Requirement Type	Heading
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- CES_Basic DID/PID Definition
- CES_CAN High Speed Specification
- CES_CAN FD Specification
- CES_DoIp Specification
- CES_ECU Identifier Definition
- CES_Flash Specification
- CES_Security Access

11.3 Referenced Documents

Req ID	379	VinFast ID		Version number	4	Requirement Type	Heading
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DTC_DID_PID range for each component.xlsx

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